

Authoring Spec — AI End-to-End Learning Track

Appendices · [_tools/AUTHORING_SPEC.md](#)

Every module/track folder in this repo follows the **same** contract. Read this before writing anything. The gold-standard reference implementation is `/Users/srinip/ai-all/01-ml-foundations/` `[01-ml-foundations]` — skim its `README.md`, `notes/01-core-concepts.md`, `code/`, `lab/EXERCISES.md` and `slides/deck.json` and match that voice, density and structure.

1. Required files

```
<track>/
├─ README.md                # the front door (see §2)
├─ notes/01-<topic>.md      # core concepts, teaching prose
├─ notes/02-<topic>.md      # deep dive / practitioner craft
├─ code/<name>.py|.ts        # 2+ RUNNABLE examples, heavily commented
├─ code/requirements.txt    # or package.json for TS
├─ lab/EXERCISES.md         # graded exercises with explicit deliverables + checks
├─ papers/PAPERS.md         # primary sources / official docs, real URLs, annotated
├─ resources/RESOURCES.md   # courses, repos, docs, communities – real URLs only
└─ slides/deck.json         # slide source (see §3) – do NOT write .pptx yourself
```

2. README.md shape

- H1 with an emoji + track name.
- A blockquote line with time estimate and prerequisites.
- 2–4 sentences on *why this matters* and where it sits in the track.
- **## Learning objectives** — 5–7 numbered, testable capabilities ("you can X"), not topics.
- **## Suggested path** — a markdown table: step | do this | file (relative link) | time.
- **## The N terms you must own** — inline-code list of key vocabulary.
- **## Exit check**  — a concrete, verifiable deliverable.

3. slides/deck.json schema

```
{
  "module": "16.1",
  "title": "Track Title",
  "subtitle": "One sharp line",
  "accent": "64748B",
  "slides": [
    {"type": "bullets", "title": "...", "bullets": ["...", "- nested sub-point"], "notes": "speaker note"},
    {"type": "table", "title": "...", "columns": ["A", "B"], "rows": [["1", "2"]], "notes": "..."},
    {"type": "quote", "title": "...", "quote": "one line", "bullets": ["..."], "notes": "..."},
    {"type": "section", "title": "...", "subtitle": "...", "notes": "..."}
  ]
}
```

Rules: 14–20 slides. Max 6 bullets/slide, max ~14 words each. Tables max 6 rows, 4 columns. **ASCII only inside deck.json** (no smart quotes, em dashes, arrows, or emoji — use →, --). Every slide gets a `notes` field with something a presenter would actually say. A title slide and an outro slide are added automatically — do not write them. Build with: `/Users/srinip/ai-all/.venv/bin/python /Users/srinip/ai-all/_tools/build_decks.py <track-folder-name>` and confirm it prints a ✓ line.

4. Quality bar — this is the part that matters

- **Teach, don't list.** Explain *why*, show the mechanism, name the failure mode. A bulleted glossary is a failure.
- **Opinionated.** Say which option you would pick and why. Name the common mistake.
- **Concrete.** Real library names, real version-era behaviour, real numbers, real commands.
- **Tables** for comparisons; **ASCII diagrams** for architectures; short code blocks inline in notes.
- **Forward/backward links** to other modules using relative paths (e.g. `../08-rag/`), with a "where this returns" table at the end of the core-concepts note.
- **Runnable code:** scripts must execute top-to-bottom and print something instructive. If a script needs network/API keys, guard it so it degrades to an offline explanation instead of crashing, and say so in a docstring.
- **No fabrication.** Every URL must be one you are confident exists. Prefer official docs, arXiv, and well-known GitHub repos. If unsure of a URL, link the project root rather than a deep path.
- Notes files: aim 150–250 lines each. Be substantive; do not pad.

5. Tone

Direct, technical, slightly opinionated, occasionally dry-funny. Write for a competent engineer who is new to this specific topic and hates being condescended to. Second person. No marketing language.

6. Building the book

`_tools/build_book.py` turns the markdown of the whole repository into one print-ready PDF plus a PDF per chapter:

```
.venv/bin/python -m pip install markdown pypdf websockets pypdfium2
.venv/bin/python _tools/build_book.py                # whole book -> book/
.venv/bin/python _tools/build_book.py 04-llm 08-rag   # a subset, for a quick look
.venv/bin/python _tools/build_book.py --help
```

Discovery is by folder (00-launchpad, 01-...-16-..., 16-.../tracks/*, 99-capstones), so a new module appears in the book automatically once it has `README.md`, `notes/`, `lab/`, `papers/` and `resources/`. A part's chapters are, in order: the README (as **Overview**), each `notes/*.md`, the lab, any markdown under `code/` (templates, prompt patterns), then `papers/PAPERS.md` + `resources/RESOURCES.md` as one **Sources** chapter. `slides/slides.md` — a generated export of the deck — and non-markdown code are deliberately not printed.